

DBMS End-Sem Question Paper 2024

C.V. Raman Global University • Computer Science & Engineering • Semester 3

SECTION 1: Relational Design & Decomposition [20 Marks]

1. Explain why 3NF is considered a pragmatic compromise over BCNF for dependency preservation.
2. Prove that a 2-attribute relation is always in BCNF.
3. Discuss Armstrong axioms for functional dependencies and prove the Transitivity rule.

SECTION 2: Concurrency Control & Recovery [20 Marks]

4. Discuss the ACID properties of transactions. How does the Write-Ahead Logging (WAL) protocol guarantee Atomicity and Durability?
5. Contrast Strict 2PL and Rigorous 2PL. How does Strict 2PL eliminate cascading rollbacks?
6. Explain the Wait-Die and Wound-Wait schemes for deadlock prevention.